TAEHYEOK JANG

Software Engineer specializing in data infrastructure and video analytics
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PROFESSIONAL EXPERIENCE

Graduate Researcher, Data Systems Lab (Advisor: Kexin Rong) 2025.01–present

- ***KeyframeQL*** (submitted to *SIGMOD `25 workshop* [Paper])
  - Designed a video query engine **for multi-modal video analytics,** enabling expressive spatio-temporal queries (e.g., "two cars crossing an intersection") on large-scale autonomous datasets (VIRAT, nuScenes, Waymo Open) and safety scenarios (EURO NCAP 2025 - crash avoidance).
  - Built **ML preprocessing pipelines** for tagging and indexing of raw videos (e.g., traffic cameras), including trajectory extraction and event/action recognition.
  - Integrated **NL2SQL** pipeline using LangChain and schema-aware IR generation, providing scalable **dataset curation** for complex traffic and driving scenarios.
  - Partnered with ***Bosch Research Labs*** to develop autonomous data query systems.

Sr. Backend Engineer, Coupang, Inc - Eats Engineering 2021.08–2024.04

(Coupang is the #1 e-commerce platform in South Korea)

- **Real-Time Data Streaming and Analytics Platform** (12M+ events/day, 2.5K+ queries/day)
  - Architected **Kafka-based ingestion pipelines** and **columnar data lake (ClickHouse, Trino)** for fleet operations data.
  - Built CDC-like producer SDK, schema registry, and idempotent ingestion for reliable data persistence.
  - Delivered **data search & analytics tooling** (Redash) for rider observability, and ML-based features.
  - Secured **KR Patent 1026779280000**.
- Designed and scaled **the Rider Platform** to support 600K+ daily orders and 60K+ concurrent riders.
- Built distributed **dispatch infrastructure** (multi-order, push/pull-based matching) to ensure high-throughput, low-latency rider-order matching across regions.

Software Engineer, LINE Corp - LINE Messaging Infrastructure 2019.01–2021.07

(LINE is a top messaging platform with 200M+ MAUs in East Asia)

- Implemented chat group features supporting 100+ concurrent members, optimizing throughput and storage efficiency with **HBase**.
- Built large-scale **data synchronization and replication system** for 120M+ users, handling social graph ingestion at 96K msg/sec peak throughput using **multi-threaded Kafka and HBase**.

Software Engineer - Republic of Korea Air Force (military service) 2016.02–2018.02

- Designed and built internal notification system with visualization and custom messaging protocol

EDUCATION

M.S. in Computer Science (*GPA 3.90/4.00)* 2024.01-2025.12 (expected)

Georgia Institute of Technology / Atlanta, GA

Coursework: *Database Internals, Advanced Operating Systems, Graduate Algorithms, Machine Learning, Deep Learning, Large-Scale & Real-Time Visual Analysis*

B.S. in Electrical and Computer Engineering 2011.03-2015.08

Seoul National University / Seoul, Republic of Korea

SKILLS

- **Data & Query Systems:** Apache Spark, Trino, MySQL, PostgreSQL**,** ClickHouse, HBase, Debezium
- **Distributed Systems:** Kafka, Airflow, Kubernetes, gRPC, CDC pipelines, event streaming
- **AI / ML**: PyTorch, NumPy, LangChain, YOLO, ByteTrack, embedding similarity
- **Programming Languages:** C, C++, Java, Python